

Khush Modi

❖ khush.modi@outlook.com ❖ (201) 273-4923 ❖ Somerset, NJ ❖ <https://www.linkedin.com/in/khush-modi>

WORK EXPERIENCE

Panva technology
Full-Stack Software Engineer Intern

May 2024 - August 2024
Pomay, CA

- Revamped legacy Java-based backend for payment processing, reducing latency by 25% using modular API design.
- Collaborated with frontend engineers to deliver a new responsive UI for fintech clients using React and RESTful services.
- Conducted code reviews and wrote unit tests, improving code reliability and maintainability by 30%.
- Integrated logging and monitoring tools, increasing backend observability and debugging efficiency.

Rutgers University
Course Assistant

May 2022 - August 2022
Camden, NJ

- Tutored 30+ students in College Algebra and Pre-Calculus, addressing individual learning needs and improving student comprehension of core mathematical concepts.
- Led weekly review workshops and one-on-one tutoring sessions, resulting in a 25% increase in student confidence as measured by feedback surveys and performance improvements.
- Contributed to a 75% improvement in student grades, with more students earning A's and B's compared to previous semesters by providing consistent, personalized support and exam preparation.

PROJECTS

Product Recommendation Engine

September 2024 - December 2024

- Developed a Python-based recommendation system that analyzed 100+ purchase records to suggest personalized products.
- Created features like "Most Popular Products" and "Your Best Picks", with a simulated 10% increase in the click-through rate (CTR).
- Evaluated performance by tracking simulated user engagement and adjusting recommendation algorithms based on real-time data.

Twitter Data Insights Tool

January 2024 - April 2024

- Extracted and processed data from 500+ Twitter users using Twitter APIs (GET users/show, GET users/lookup, GET followers/ids), analyzing user engagement across devices (iPhone vs. Android).
- Engineered a simulation application that tracked device-specific tweet behaviors, identifying and quantifying key differences in tweet volume, retweets, and likes between platforms.
- Uncovered trends in tweet engagement, revealing how platform-specific factors influenced content interaction and user behavior.
- Developed a custom query interface, enabling real-time data extraction and trend analysis, resulting in a 40% increase in data processing.

Air-Hub – Flight Booking Web Platform

September 2023 - December 2023

- Developed a full-stack flight booking platform with role-based access (Customer, Admin, Supervisor) using JSP and SQL, supporting 100+ users with secure login and role-specific functionalities.
- Implemented real-time flight availability checks and an e-commerce-style multi-item cart, simulating over 200+ flight bookings.
- Developed secure user authentication and session management, ensuring robust access control and data protection for all roles.

Signal Light Intersection Simulator

January 2023 - April 2023

- Designed and developed a traffic signal simulation for a 4-way intersection, testing multiple signal light combinations (red/yellow/green).
- Created timer-based control for signal transitions and simulated 10+ traffic flow scenarios, improving the understanding of signal timing.
- Validated the system's performance through automated testing, ensuring accurate behavior under various traffic conditions.

EDUCATION

Rutgers University-New Brunswick

January 2025

Bachelor of Science in Computer Science (GPA: 3.6/4.0)

- Member: USACS (Computer Science student organization), RUOC (Rutgers Outdoors Club)
- Relevant Coursework: Data Structures, Algorithms, OOP, Computer Architecture, Software Engineering, AI, Data Science

SKILLS

- **Programming Languages:** Python, Java, JavaScript, TypeScript, C, C++, C#, SQL
- **Frameworks & Libraries:** React, Django, Node.js, Pandas, NumPy, scikit-learn
- **Web Technologies:** HTML, CSS, RESTful APIs, JSON
- **Databases & Cloud:** PostgreSQL, MySQL, SQLite, AWS (EC2, S3, RDS – basic)
- **DevOps & Tools:** Git, GitHub, Docker, GitHub Actions, Linux, VS Code

CERTIFICATIONS

- **Cisco:** Linux Unhatched • **IBM:** Data Analysis Using Python • **Hacker Rank:** C# Programming